Leveraging Structure for Tractable Reasoning in Formal Argumentation

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Abstract

Argumentation is a prominent explainable approach to addressing non-monotonic reasoning tasks, with applications in e-government, healthcare, and legal contexts. The intractability of many argumentative reasoning tasks led to a growing interest in dedicated efficient algorithms. This thesis addresses these challenging reasoning tasks by focusing on the structure of argumentation frameworks. We employ two primary approaches: i. developing tree decomposition-based algorithms, and ii. reducing complexity through instance preprocessing. Our results have been promising so far, encouraging further investigation of possible extensions of our algorithms to different argumentation formalisms.

1 Introduction

Argumentation can be described as a long-lived decision-making mechanism in which the participants make, support, or counter certain claims, so-called arguments towards settling a conflict of opinions. Formal and computational approaches to argumentation emerged in the field of Artificial Intelligence (AI) in the 80s, as one way to formalise non-monotonic reasoning (Baroni et al. 2018). Since argumentation is by nature explainable, and it provides exact outcomes, its applications include the legal, healthcare, and e-government domains (Atkinson et al. 2017; Bench-Capon, Prakken, and Sartor 2009). As a consequence, the interest in efficient models for argumentation has increased over the years, and there is one bi-annual international competition on computational models of argumentation (ICCMA) (Järvisalo, Lehtonen, and Niskanen 2023; Bistarelli et al. 2021; Thimm and Villata 2017; Lagniez et al. 2020; Gaggl et al. 2018), that aims at comparing argumentation solvers in terms of running time. Several more applications of argumentation are discussed by Simari and Rahwan 2009.

In formal argumentation, a major distinction is made between abstract argumentation (Dung 1995) and structured argumentation (Besnard et al. 2014), i.e. between formalisms that only consider arguments as atomic, and formalisms that support the expression of arguments' internal structure. In addition, another branch of works tries to model more complex argumentation scenarios, which include some sort of uncertainty. One prominent approach is to assign probabilities to arguments and attacks in so-called probabilistic abstract argumentation frameworks (PAFs) (Hunter et al. 2021; Hunter and Thimm 2017), where, intuitively, an argument (or attack) having probability 1 is believed surely, whereas an argument (or attack) having probability less than one is only believed to the specified degree. Argumentative reasoning with uncertainty and models thereof have recently benefited from an increase of interest.

In formal argumentation, decision making is generally achieved by solving reasoning tasks defined in terms of acceptability of arguments according to some criteria, so-called *semantics*. A set of arguments is said to be *admissible* if it defends itself against all external attacks. Such set is also *preferred*, if it is also maximal. To compute the acceptability of arguments, and in general reasoning tasks, one operates over argumentation frameworks which are often graph representations of an argumentation instance. The computability of many reasoning tasks is predominantly intractable (Dvořák and Dunne 2018; Čyras et al. 2018), and significant research effort is dedicated to overcome their complexity. Dunne 2007 has categorised available approaches to tackle said complexity into *i.* works that consider certain graph-theoretic properties of a problem, and *ii.* works that consider structural aspects of an argumentation framework, such as arguments' structure and the attack relationship.

In line with approach *i.*, one well-studied graph property is *tree-width* (Bodlaender 1993), informally indicating the closeness of a graph to a tree. In particular, Courcelle's theoretical results (Courcelle 1990) guarantee that, in certain conditions, fixed parameter tractable (FPT) algorithms exist for problem instances with a bounded tree-width. A parameterised (by k) problem $\langle I, k\rangle$ is said to be FPT if it can be solved by an algorithm with running time $O(f(k) \cdot |I|^c)$, where f is any function, $|I|$ is the instance size, and c is a constant. In particular, due to Courcelle (Courcelle 1990), we know about the existence of FPT algorithms parameterised by tree-width for a given task, if the task can be encoded in Monadic Second Order logic (MSO).

With regard to approach ii. (considering structural aspects of an argumentation framework), one can, in general, preprocess problem instances to reduce their complexity, and to proceed with algorithms development thereafter.

2 Motivation

While some approaches are available for abstract argumentation (Dvořák, Pichler, and Woltran 2012), little work has been conducted in other settings, e.g. in structured argumentation and in probabilistic argumentation.

It is in general not trivial to lift results obtained on abstract argumentation formalisms to structured argumentation (Lehtonen, Wallner, and Järvisalo 2020; Lehtonen, Wallner, and Järvisalo 2021). In terms of complexity, credulous acceptance of arguments under several semantics (admissible, preferred, stable, and complete) is intractable for the $flat$ fragment of assumption-based argumentation (ABA) (Bondarenko et al. 1997; Čyras et al. 2018). Simultaneously, it is commonly agreed that structured argumentation frameworks are better suited for real-world applications, strongly motivating further research towards efficient approaches in structured argumentation.

In regards to probabilistic argumentation frameworks, we are not aware of many works tackling the higher complexity of reasoning. When considering the *constellation approach* (Li, Oren, and Norman 2011) and assuming independence among arguments, the tasks of computing the probability of a given set of arguments being a σ-extension, and credulous acceptance of an argument have been shown to be intractable (Fazzinga, Flesca, and Parisi 2015; Fazzinga, Flesca, and Furfaro 2018; Fazzinga, Flesca, and Furfaro 2019; Alfano et al. 2023).

There have only been few approaches addressing hard problems in structured argumentation frameworks and in probabilistic abstract argumentation frameworks so far. In specific, to the best of our knowledge, there were, prior to our work, no available approaches utilising tree-width for reasoning in ABA, and in general, no available working systems tailored towards performance were developed. In addition, this gap extends over different possible fragments, or more expressive variations of argumentation models (e.g. bipolar probabilistic argumentation frameworks).

In general, our work contributes to the corpus of limited computational approaches available, setting the stage and encouraging other researchers in the field to tackle efficiency in hard argumentative reasoning tasks, or to investigate further structural properties that can be exploited for tractable reasoning.

3 Current Progress

We started our work by focusing on the well-known structural property of graphs, tree-width, and tackled the prominent structured argumentation model ABA. We utilised the framework provided by Courcelle's theorem (Courcelle 1990) to exploit tree-width towards efficient reasoning in ABA. In particular, we developed a fixed-parameter tractable (FPT) approach, parameterised by tree-width (Popescu and Wallner 2023). Next, we moved on to probabilistic abstract argumentation frameworks, and worked on FPT algorithms for computing the probability of a set of arguments being a σ-extension in the constellation approach (Popescu and Wallner 2024). In both works, we have implemented our algorithms in working prototypes in order to evaluate their performance. We used the dynamic programming, ASP-based framework D-FLAT (Abseher et al. 2014; Abseher et al. 2018) for the instantiation of our algorithms for the *flat*-ABA fragment, while we implemented from scratch a Python prototype for the work on probabilistic argumentation that we conducted in (Popescu and Wallner 2024). We have spent considerable time in assessing whether D-FLAT would be a reasonable choice in the probabilistic setting (in the *constellation approach* specifically), but we encountered barriers in terms of precision. Our work on assumption-based argumentation was published in the proceedings of JELIA23, while our recent work on probabilistic argumentation was accepted in KR2024.

Nonetheless, there are still unaddressed research gaps, for example when it comes to computing the probability of an argument being accepted under a given semantics in PAFs, as well as when considering both problems in variants of PAFs, such as bipolar abstract argumentation frameworks (BAFs) (Amgoud et al. 2008). We are not aware of many approaches in these directions, with the exception of the work of Fazzinga et al. 2018, who proposed an algorithm for computing the probability of a given set of arguments being a σ-extension in BAFs.

4 Plan and Future Works

Based on the overall positive results obtained so far, we want to further investigate tree decomposition-based algorithms and pre-processing techniques for hard reasoning tasks in different argumentation formalisms. Here we briefly outline a couple of directions that we plan to explore as extensions of our work so far.

A well-studied variation of AFs are the so-called bipolar argumentation frameworks, in which, in addition to attack relations, support relations between arguments can be used to express further interactions. We would like work on a possible extension and evaluation of the algorithms we proposed in (Popescu and Wallner 2024) to probabilistic bipolar argumentation frameworks.

In (Popescu and Wallner 2024) we have focused on computing the probability of a set of arguments being a σ-extension. We would like to study how one could extend the proposed algorithm to compute the acceptance probability of an argument under a given semantics.

Moreover, we would like to further contribute to tackling hardness of reasoning tasks in structured argumentation (ABA), by, e.g., investigating an extension of our work on *flat*-ABA to the full formalism.

Another possibility consists of extending our findings on the benefits of FPT algorithms to other structured argumentation models, such as ASPIC$^+$ (Modgil and Prakken 2013; Modgil and Prakken 2018), probabilistic assumption-based argumentation (PABA) (Dung and Thang 2010), and in the Defeasible Logic Programming (DeLP) formalism (García and Simari 2004; García and Simari 2018). Finally, there are possibilities to extend our work in terms of considering other structural properties of argumentation instances.

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